

UAV MOTION PLANNING HANDOUT

Why Local Planning Methods Can Get Stuck

Nonconvex collision optimization, KKT points, and potential-field traps

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Claims supported by slide 14. Obstacle avoidance generally makes the feasible set nonconvex. Consequently, a local nonlinear optimizer normally guarantees only a nearby stationary/KKT point, and the initial trajectory can select its basin. For a reactive potential field, gradient descent decreases energy, but it stops at *every* critical point; a stable critical point need not be the goal.

1 Convexity and the obstruction created by obstacles

A set $S \subset \mathbb{R}^d$ is convex when

$$x, y \in S \implies (1 - \theta)x + \theta y \in S \quad \forall \theta \in [0, 1].$$

Let a circular obstacle be

$$\mathcal{O} = \{(x, y) : x^2 + y^2 \leq r^2\}, \quad \mathcal{F} = \mathbb{R}^2 \setminus \mathcal{O}.$$

The two feasible points $z_+ = (0, r + \varepsilon)$ and $z_- = (0, -r - \varepsilon)$ satisfy $z_{\pm} \in \mathcal{F}$, but their midpoint is

$$\frac{1}{2}z_+ + \frac{1}{2}z_- = (0, 0) \in \mathcal{O}.$$

Therefore $\mathcal{F}$ is not convex. The same argument applies in trajectory space: a safe path above an obstacle and a safe path below it can have a pointwise average that intersects the obstacle.

2 Homotopy choices create separate basins

Two paths with common endpoints are in the same homotopy class if one can be continuously deformed into the other while remaining collision-free. Paths passing on opposite sides of a blocking obstacle are often in distinct classes. A local update

$$\xi_{k+1} = \xi_k + \Delta \xi_k$$

that preserves feasibility and uses small steps usually cannot cross the infeasible barrier separating those classes. Thus initialization is not cosmetic: it chooses which connected neighborhood or basin the solver explores.

This does not imply every obstacle problem has multiple local minima. It proves that the convex guarantee is lost and supplies a mechanism by which distinct locally optimal routes arise.

3 What KKT stationarity actually guarantees

Consider

$$\min_x f(x) \quad \text{s.t.} \quad h(x) = 0, \quad g(x) \leq 0.$$

At a regular local optimum x^* , there may exist multipliers λ^* and $\mu^* \geq 0$ satisfying

$$\begin{aligned} \nabla f(x^*) + Dh(x^*)^T \lambda^* + Dg(x^*)^T \mu^* &= 0, \\ h(x^*) &= 0, \quad \text{quad}g(x^*) \leq 0, \\ \mu_j^* g_j(x^*) &= 0. \end{aligned}$$

These are necessary first-order conditions under a constraint qualification. In a nonconvex problem they are not sufficient for global optimality and may even describe maxima or saddles.

Explicit local-but-not-global minimum

Let

$$f(x) = (x^2 - 1)^2 + \varepsilon x, \quad 0 < \varepsilon \ll 1.$$

Then

$$f'(x) = 4x(x^2 - 1) + \varepsilon, \quad f''(x) = 12x^2 - 4.$$

For small ε , there are stationary points near $x = -1, 0, 1$. The outer two have $f'' > 0$, so both are strict local minima, but the tilt εx makes the minimum near -1 lower than the one near $+1$. A local solver started in the right basin can converge to a valid KKT point that is not globally best.

4 Gradient potential fields

Let $U : \mathbb{R}^d \rightarrow \mathbb{R}$ be a differentiable artificial potential and choose the command

$$\dot{x} = -K\nabla U(x), \quad K = K^T \succ 0.$$

Along a trajectory,

$$\dot{U}(x) = \nabla U(x)^T \dot{x} = -\nabla U(x)^T K \nabla U(x) \leq -\lambda_{\min}(K) \|\nabla U(x)\|^2 \leq 0.$$

This is the precise useful statement: energy never increases, and it decreases strictly wherever $\nabla U \neq 0$. But equilibrium means

$$\dot{x} = 0 \iff \nabla U(x) = 0,$$

which includes the goal, spurious local minima, maxima, and saddles.

5 Stability is local and determined by the Hessian

At a critical point x_e , linearize with $e = x - x_e$:

$$\dot{e} \approx -KH_e e, \quad H_e = \nabla^2 U(x_e).$$

If $H_e \succ 0$, then x_e is a strict local minimum and the linearized dynamics are attracting. If H_e has a negative eigenvalue, the point is unstable in at least one direction. Thus the controller cannot distinguish a goal minimum from an unintended minimum using the local gradient alone.

For $K = \alpha I$ and an m -strongly convex, L -smooth potential in a neighborhood of the desired goal, gradient descent with a suitable step size converges rapidly there. The qualifier “in a neighborhood” is essential: global strong convexity is usually destroyed by obstacle repulsion.

6 A complete analytic trap example

Consider

$$U(x, y) = (x^2 - 1)^2 + y^2.$$

Its gradient and Hessian are

$$\nabla U = \begin{bmatrix} 4x(x^2 - 1) \\ 2y \end{bmatrix}, \quad \nabla^2 U = \begin{bmatrix} 12x^2 - 4 & 0 \\ 0 & 2 \end{bmatrix}.$$

The critical points are $(1, 0)$, $(-1, 0)$, and $(0, 0)$. At $x = \pm 1$,

$$\nabla^2 U = \text{diag}(8, 2) \succ 0,$$

so both are strict stable minima under $\dot{z} = -\nabla U(z)$. At $(0, 0)$ the Hessian is $\text{diag}(-4, 2)$, hence it is a saddle. If $(1, 0)$ is declared the goal, then $(-1, 0)$ is a mathematically explicit stable trap with the same minimum energy.

The x dynamics are

$$\dot{x} = -4x(x^2 - 1) = 4x(1 - x^2).$$

If $x(0) > 0$, the trajectory approaches $+1$; if $x(0) < 0$, it approaches -1 . The saddle's stable boundary $x = 0$ separates the two basins. This makes “the initial guess selects the basin” exact rather than metaphorical.

7 Attractive plus repulsive potentials

A common construction is

$$U(q) = U_{\text{att}}(q) + \sum_j U_{\text{rep},j}(q), \quad U_{\text{att}}(q) = \frac{1}{2}k_a\|q - q_g\|^2.$$

For obstacle distance $\rho_j(q)$ and influence distance ρ_0 ,

$$U_{\text{rep},j}(q) = \begin{cases} \frac{1}{2}k_r \left(\frac{1}{\rho_j(q)} - \frac{1}{\rho_0} \right)^2, & 0 < \rho_j(q) < \rho_0, \\ 0, & \rho_j(q) \geq \rho_0. \end{cases}$$

A trap occurs wherever attractive and repulsive gradients cancel:

$$k_a(q - q_g) + \sum_j \nabla U_{\text{rep},j}(q) = 0,$$

even though $q \neq q_g$. If the total Hessian is positive definite there, the cancellation point is a stable local minimum.

8 Discrete descent and its limitation

For $x_{k+1} = x_k - \alpha \nabla U(x_k)$ and an L -smooth U , the descent lemma gives

$$U(x_{k+1}) \leq U(x_k) - \alpha \left(1 - \frac{L\alpha}{2} \right) \|\nabla U(x_k)\|^2.$$

Thus $0 < \alpha < 2/L$ guarantees decrease. This still establishes only convergence toward the set of critical points, not necessarily toward the desired goal or global minimum.

9 What remedies can and cannot do

Remedy	Mathematical role
Multiple initial trajectories	Searches more basins; no universal finite global guarantee.
Global graph/sampling seed	Supplies distinct homotopy classes to a local optimizer.
Convex safe corridors	Replaces raw obstacle avoidance with convex subproblems.
Random perturbation/escape	Can leave some saddles or shallow minima; parameters matter.
Navigation functions	Under restrictive workspace assumptions, design a unique minimum at the goal.
Global optimization	Can certify more, but usually at substantially higher cost.

10 Takeaway

The two negative claims follow from two short chains:

obstacle complement nonconvex $\Rightarrow$ separate route basins $\Rightarrow$ local KKT point need not be global,

$$\dot{x} = -K\nabla U \Rightarrow \dot{U} \leq 0, \quad \text{but} \quad \nabla U = 0 \text{ at every critical point.}$$

Local methods are valuable because their updates can be fast and smooth. Their limitation is not that they always fail; it is that descent and stationarity alone do not certify global route choice or arrival at the intended goal.